Milestone 1: Formula 1 Podium Prediction

Analytical report. Unit of analysis: one driver entering one race. Target: podium if `positionOrder ≤ 3`. Prediction cutoff: after qualifying, once the grid is fixed, and before the race starts.

1. Front-row starts and podium conversion

Question. Which circuits convert a front-row start into a podium most often, and does that ranking change after the 2014 hybrid era? Qualifying position and grid are kept separate, because penalties can move a car back after qualifying.

Grain and missingness. Each row is one cleaned driver–race (shared-drive duplicates removed; Indianapolis 500 rounds excluded). A front-row start is position 1 or 2. Circuits need at least 8 such starts to be ranked. Grid is available for essentially every start. Qualifying position is missing for most races before the mid-1990s, so the qualifying ranking uses a smaller, more modern set of rows.

Cutoff. Grid and qualifying position are known before lights out. The podium label is the outcome we measure; it is not used as a feature.

Answer. Conversion is high in both eras, and the circuit ranking changes after 2014. On the grid definition, the 2014–2024 leaders (minimum sample applied in the notebook) are Suzuka (16/18, 0.889), Zandvoort (7/8, 0.875), Jeddah (7/8, 0.875), Yas Marina (19/22, 0.864), and Monaco (17/20, 0.850). Zandvoort and Jeddah rest on only eight front-row starts, so those two ranks are less stable than Suzuka, Yas Marina, or Monaco. The pre-2014 panel is led by a different set of circuits, including Nevers Magny-Cours, Istanbul Park, Aintree, and Dijon-Prenois. The qualifying-front-row panels are similar but not identical to the grid panels, which is the expected effect of grid penalties. Figure 1 shows all four rankings.

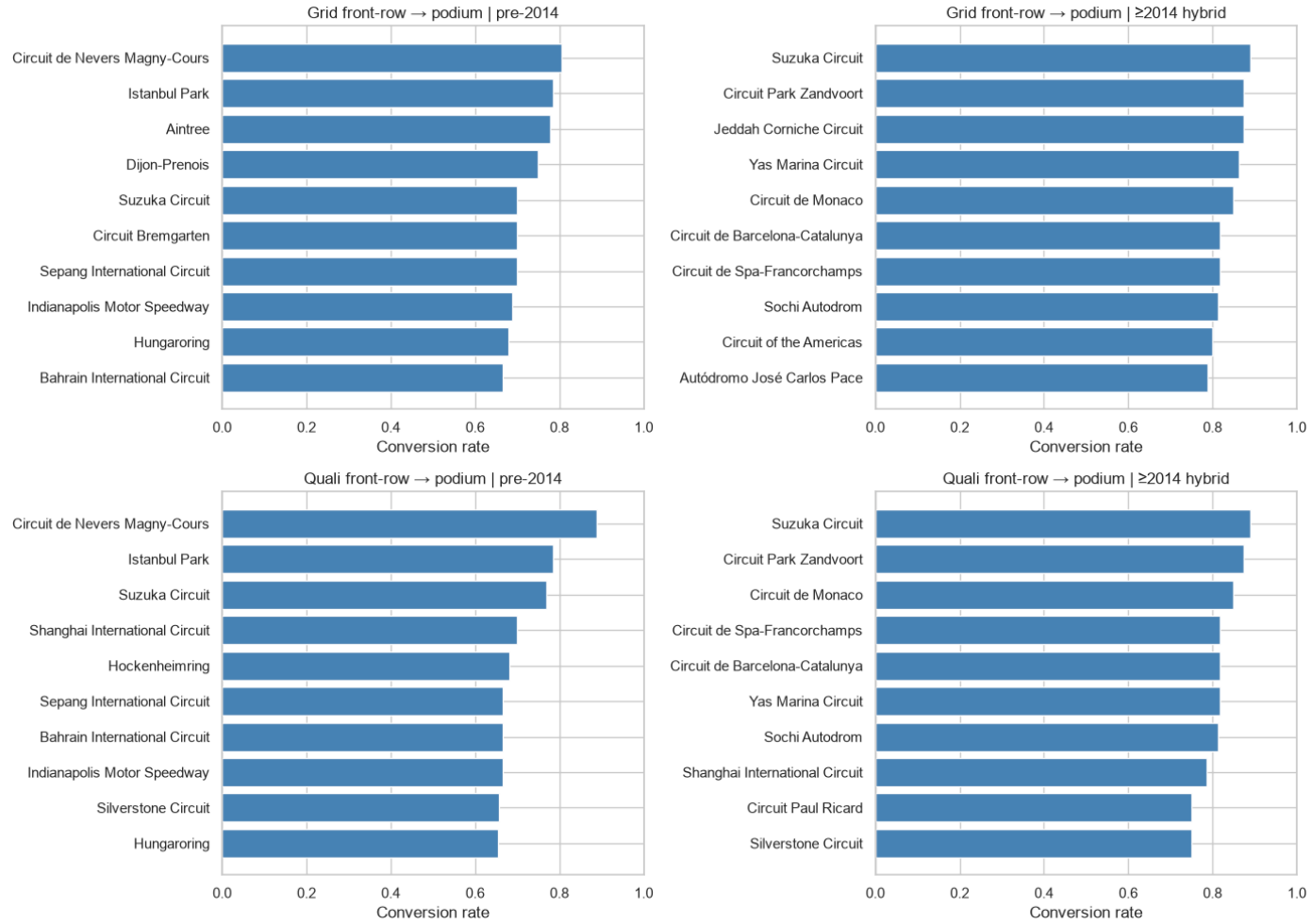


Figure 1. Podium conversion among front-row starts, split by era and by whether the front row is defined from the grid or from qualifying position.

2. Home-country podium rate, controlling for grid

Question. Do drivers podium more often at a circuit in their home country than elsewhere, after controlling for starting position?

Grain and missingness. One cleaned driver–race row. Home means the driver's nationality maps to the circuit's country (for example British → UK, American → USA). Some nationalities stay unmapped (Chilean, Liechtensteiner, and a few others) and therefore never count as home. Grid position 0 (pit-lane start) is left out of the bins. Indianapolis rounds are already excluded, so they do not inflate the American home-race count.

Cutoff. Nationality, circuit country, and grid are known before the race. This comparison measures association. It does not show that racing at home causes a podium.

Answer. Starting position dominates. Inside the same grid bin, the home gap is small, and it only favours the home driver near the front.

Grid bin	Away podium rate	Home podium rate
1–3	0.555	0.572
4–6	0.261	0.271
7–10	0.093	0.074
11–15	0.030	0.027
16+	0.007	0.003

From the front two bins the home rate is about 1–2 percentage points higher. From 7th backward the away rate is slightly higher. A raw home-versus-away comparison without grid bins would overstate a home effect, because much of the difference is starting position. Figure 2 is the controlled comparison.

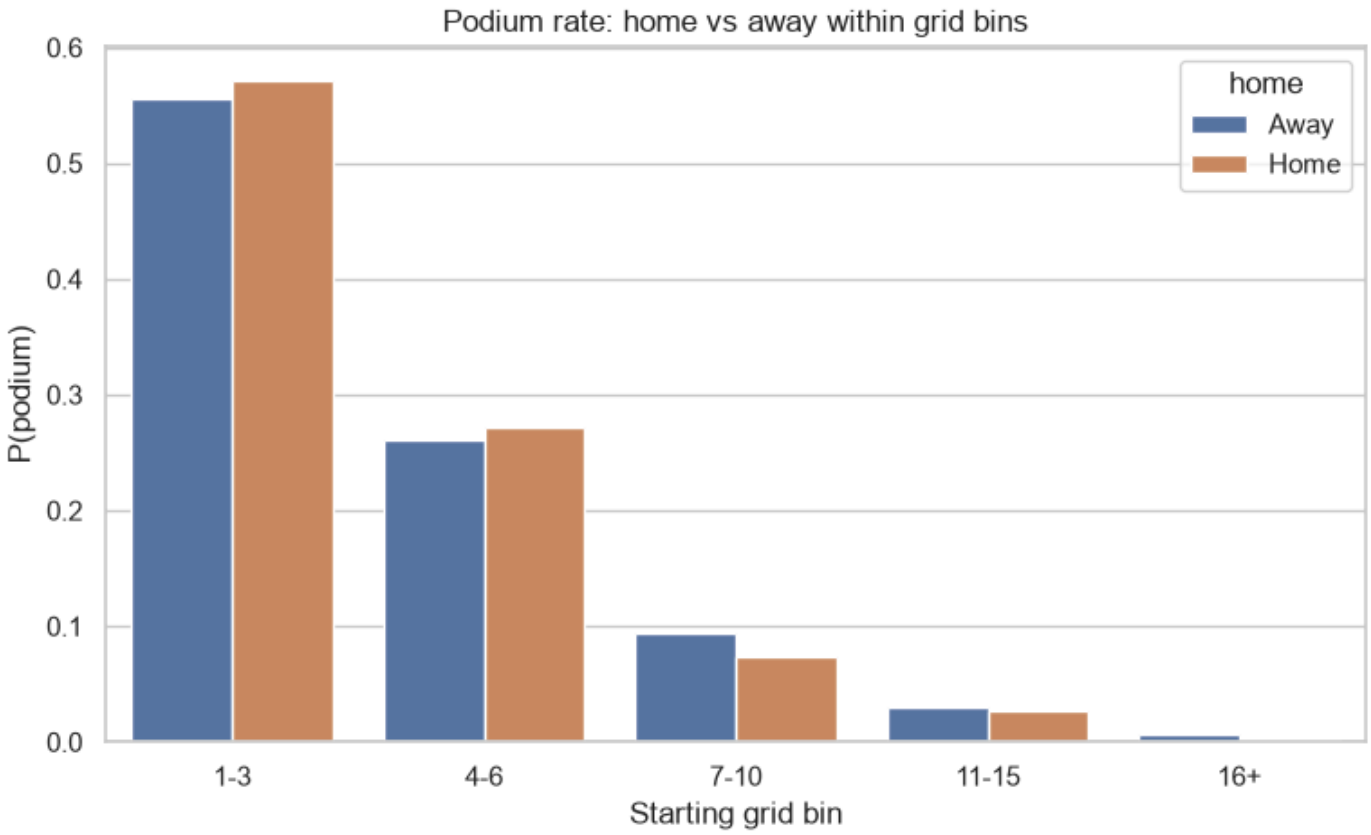


Figure 2. Podium rate for home and away starts inside the same grid bin.

3. Mechanical retirement rates, 2014–2021 versus 2022–2024

Question. Which constructors have the highest mechanical-retirement rate in the turbo-hybrid era (2014–2021) compared with the ground-effect era (2022–2024)?

Definition. A mechanical retirement is a start whose status is a component failure (engine, gearbox, hydraulics, power unit, and related codes: 73 of 139 status labels). Accidents, collisions, driver errors,

disqualifications, and failures to qualify are excluded. The mapping is documented in the notebook.

Grain and missingness. Rate = mechanical retirements / starts for that constructor in that era. Constructors with fewer than 20 starts in an era are dropped so a handful of failures cannot top the list. Status is post-race, so it is valid for this question and forbidden as a model feature.

Answer. The level of the rate falls from the first era to the second, and the constructors at the top change because the entry lists change.

Era	Constructor	Starts	Mechanical	Rate
2014–2021	Caterham	34	8	0.235
2014–2021	Lotus F1	76	15	0.197
2014–2021	Toro Rosso	242	40	0.165
2014–2021	Renault	200	30	0.150
2014–2021	McLaren	320	46	0.144
2022–2024	Alfa Romeo	88	11	0.125
2022–2024	Alpine F1 Team	136	15	0.110
2022–2024	AlphaTauri	88	7	0.080
2022–2024	Ferrari	136	9	0.066
2022–2024	Williams	135	8	0.059

Caterham's 0.235 rate is the highest in 2014–2021, on only 34 starts at the back of the grid before the team left. Among teams with a long hybrid-era sample, Toro Rosso (0.165) and Renault (0.150) are the highest. In 2022–2024 the highest rates are Alfa Romeo (0.125) and Alpine (0.110), and even those sit below the hybrid-era leaders. Figure 3 shows the top twelve constructors in each era.

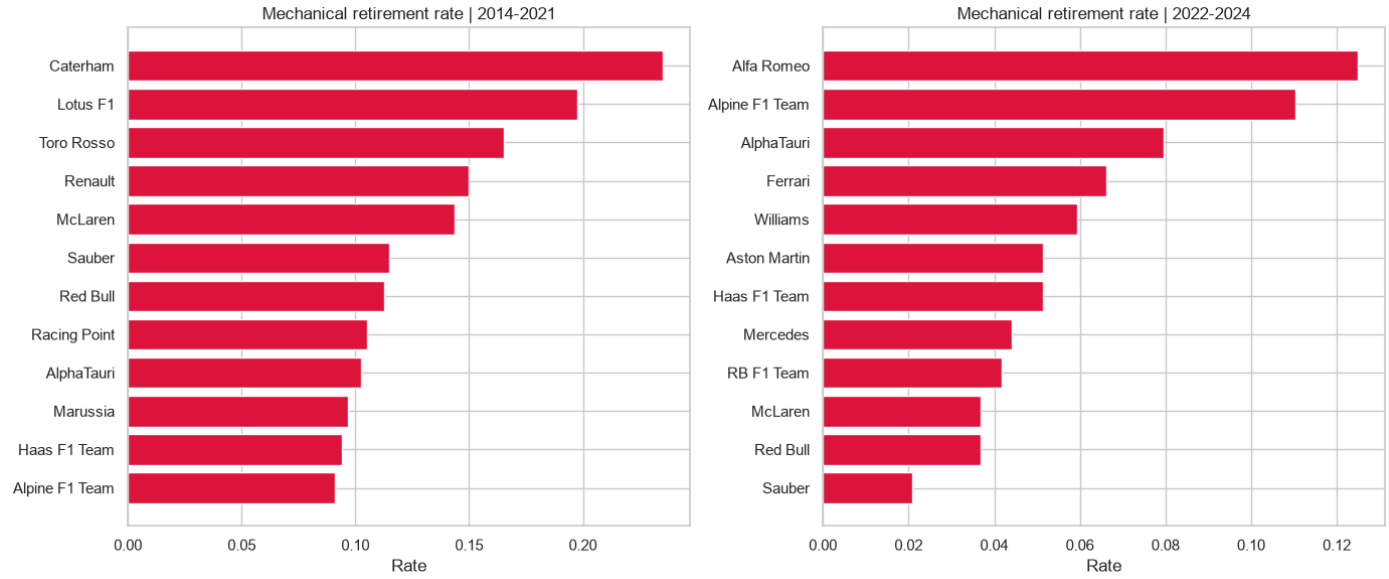


Figure 3. Mechanical retirement rate by constructor, 2014–2021 and 2022–2024. Only constructors with at least 20 starts in the era are shown.

4. Selected features and the pre-race cutoff

The label is podium versus not. Current-race finish position, points, laps, status, pit stops, and fastest lap are excluded. They are only known after the race. Sprint results are also excluded: the sprint format changed between 2021 and 2024, and using them would mix different meanings of “before the Grand Prix.”

Each kept feature is known at the cutoff, or is computed only from earlier races.

Group	Features	Why it is valid before the race
Grid	`grid_feat`	The starting grid is fixed before lights out. A grid of 0 (pit lane) is recoded to 25 so it is not treated as pole.
Qualifying	`quali_position`, `best_quali_sec`	Known once qualifying ends. When the qualifying row is missing, position is filled from the grid, which is the pre-race fallback.
Form	Driver and constructor podium rate, average finish, starts so far, driver podiums in the last five races	Each series is shifted so the current race is excluded. Only previous starts enter the feature.
Standings	Driver and constructor points and championship position	Taken from the previous race of the season, not from the standings published after this race.
Context	Year, round, circuit id, constructor id	All four are known before the start. Constructor id follows the dataset's identity, so a rename starts a new history.

Figure 4 is the pre-training check. Recent driver podium form correlates most strongly with the label (+0.48 for the last five races, +0.41 for the career-to-date rate before this race). Grid and qualifying position correlate at about −0.44: a smaller position number means a better start and a higher podium chance. Constructor podium form is +0.37. These signs match the sporting story, which is why the groups were kept. Ablation on the validation set later showed that removing grid, standings, or qualifying lowered PR-AUC, while removing the context group did not.

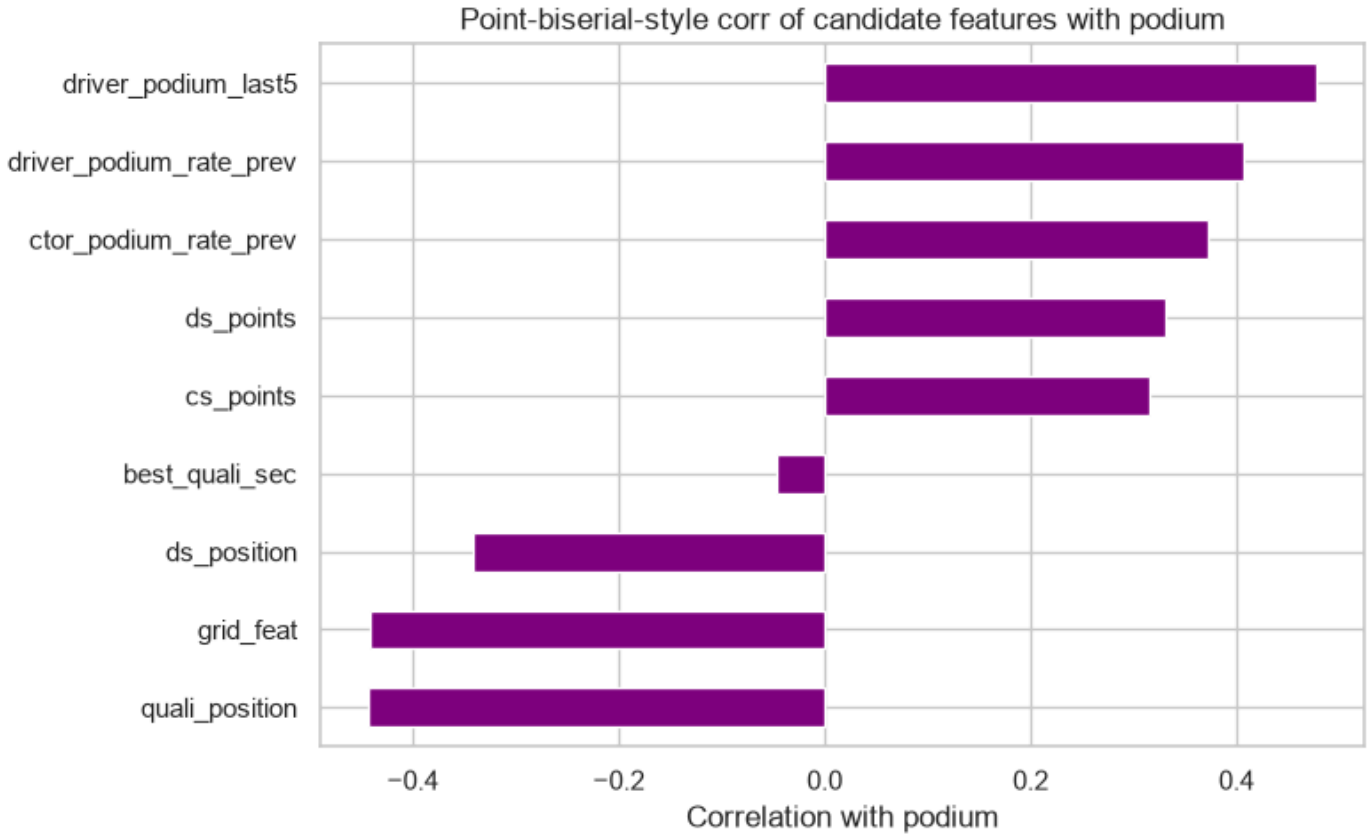


Figure 4. Correlation of candidate features with the podium label. Negative values for grid and qualifying position mean a better (lower) starting place goes with a higher podium chance.

A feature the model relies on is not a cause of podium finishes. Permutation importance and SHAP describe the fitted model, not a causal effect of grid position or team form.

5. Known limitations

- **Qualifying-coverage drift.** Qualifying rows are rare before the mid-1990s and become routine later (Figure 5). Qualifying features are therefore mostly a modern-era signal. Filling a missing qualifying position from the grid avoids dropping early seasons, and it also makes the two features partly redundant in those years.
- **Era shift.** Car rules, reliability, and which teams are fast change sharply (hybrid 2014–2021, ground effect 2022–2024). The split is chronological — train through 2019, validation 2020–2021, test 2022 onward — so the test score is an era-shift score, not a random-split score. That split reduces leakage of future rules into training. It does not remove the shift.
- **Constructor identity.** A rename is a new `constructorId`, so form and standings reset even when the people and the factory continue.
- **Cleaning choices.** Keeping the best `positionOrder` when a driver has two rows in one race, and dropping the Indianapolis 500, changes historical counts. Both rules are documented; another reasonable rule would move the early-era rows slightly.

- **Small samples in the rankings.** A circuit or constructor can rank first on a few dozen starts (Zandvoort and Jeddah front rows; Caterham mechanical rate). Those ranks are real in the table and unstable as estimates.
- **Home-country map.** Nationality strings do not match circuit country strings without a hand-built dictionary. Unmapped drivers never count as home, so the home sample is incomplete.
- **Explanations are not causes.** Global permutation importance, SHAP, and local LIME say which inputs the model uses for a prediction. They do not prove that changing that input would change the race result.

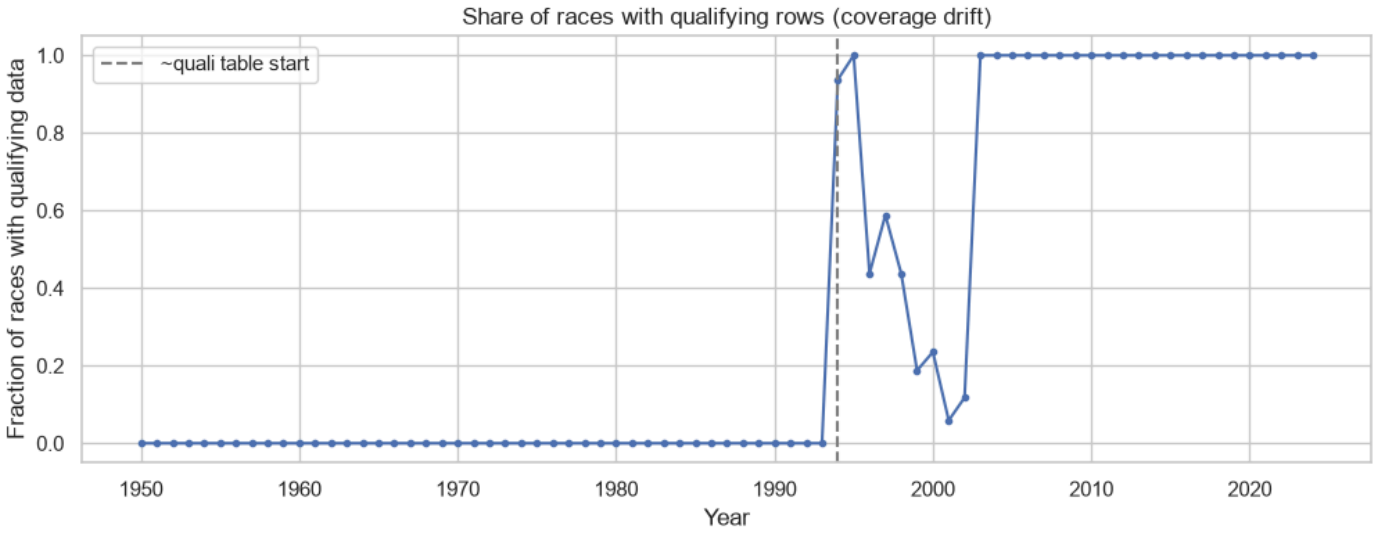


Figure 5. Share of races that have at least one qualifying row, by season. Coverage is the reason qualifying features are unreliable in early championships.

6. How the models were scored

Podiums are the minority class (about 12.8% of cleaned rows, and about three cars per race). The primary metric is therefore PR-AUC. The secondary metrics are F1, with the probability threshold chosen only on the validation set, and ROC-AUC. Median imputation followed by scaling was kept over zero-imputation followed by scaling (validation PR-AUC 0.729 versus 0.724 on logistic regression).

On validation PR-AUC the shallow network is highest (0.738). On the unseen 2022–2024 test set, histogram gradient boosting is highest (PR-AUC 0.712, F1 0.674). The validation winner does not automatically win the next rules era. Figure 6 compares validation and test.

Model	Val PR-AUC	Val F1	Val ROC-AUC	Test PR-AUC	Test F1	Test ROC-AUC
Shallow NN	0.738	0.708	0.924	0.676	0.613	0.929
HistGB	0.732	0.719	0.924	0.712	0.674	0.935
Logistic regression	0.729	0.707	0.925	0.680	0.657	0.932
Random forest	0.697	0.715	0.926	0.701	0.655	0.934

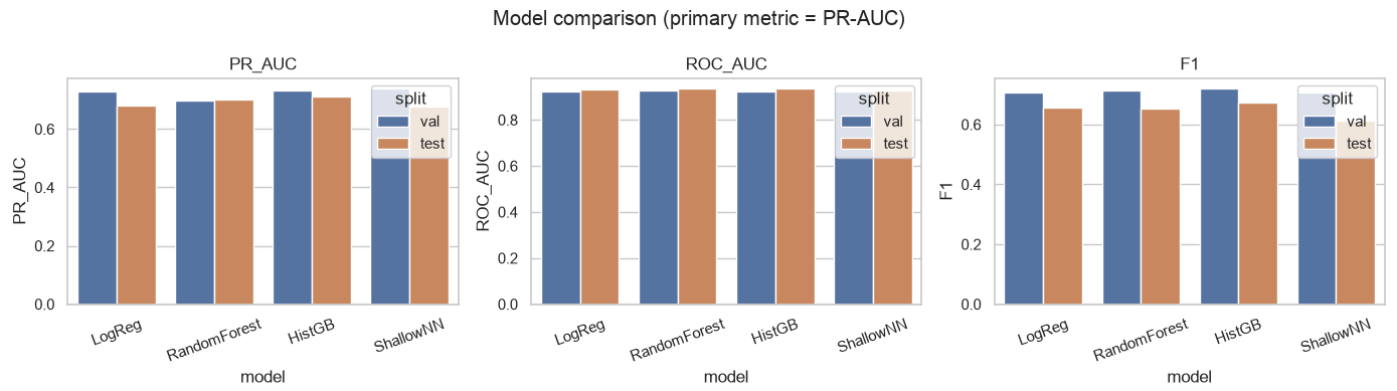


Figure 6. PR-AUC, ROC-AUC, and F1 on validation and on the held-out test seasons. Thresholds were chosen on validation only.

7. Explanations, and what they are not

Global explanations use the random forest, because its tree structure has a direct SHAP explainer. Permutation importance shuffles one validation column at a time and records the drop in PR-AUC. Grid position matters most (about 0.08), then qualifying position (about 0.04), then constructor podium form and the driver's championship position entering the race. The SHAP summary agrees: a better starting position pushes predicted podium probability up, and a worse one pushes it down.

The local plots are a single validation row. The SHAP waterfall and the LIME bar both answer "why this row?", not "how does the model behave in general?". LIME fits a simple model in a neighbourhood of that row. It can disagree in detail with SHAP because it is a local approximation.

These plots are not causal. A feature the model relies on is not proof that the feature causes podium finishes. Grid position is a strong input to this model. That does not mean moving a car one place up the grid would create a podium.

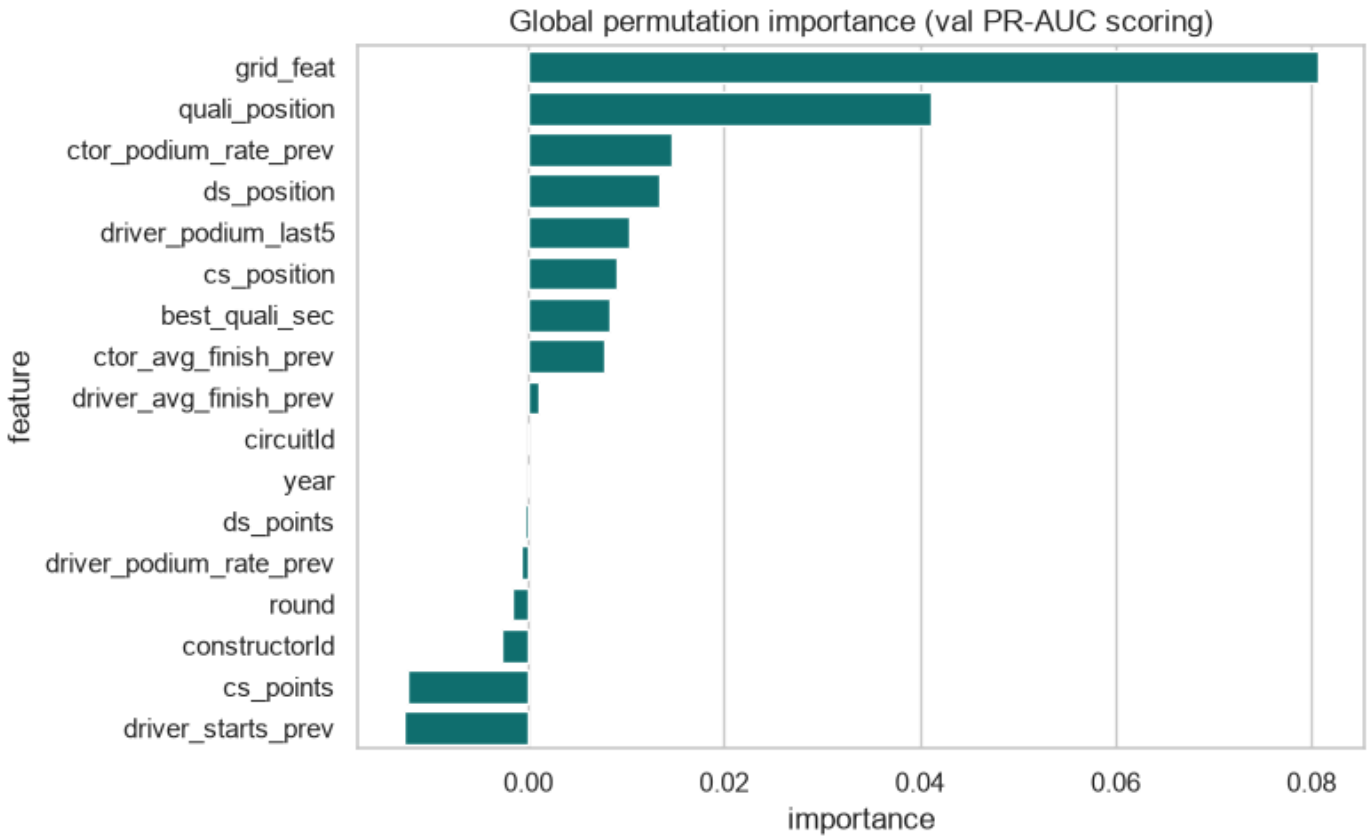


Figure 7. Permutation importance on validation PR-AUC.

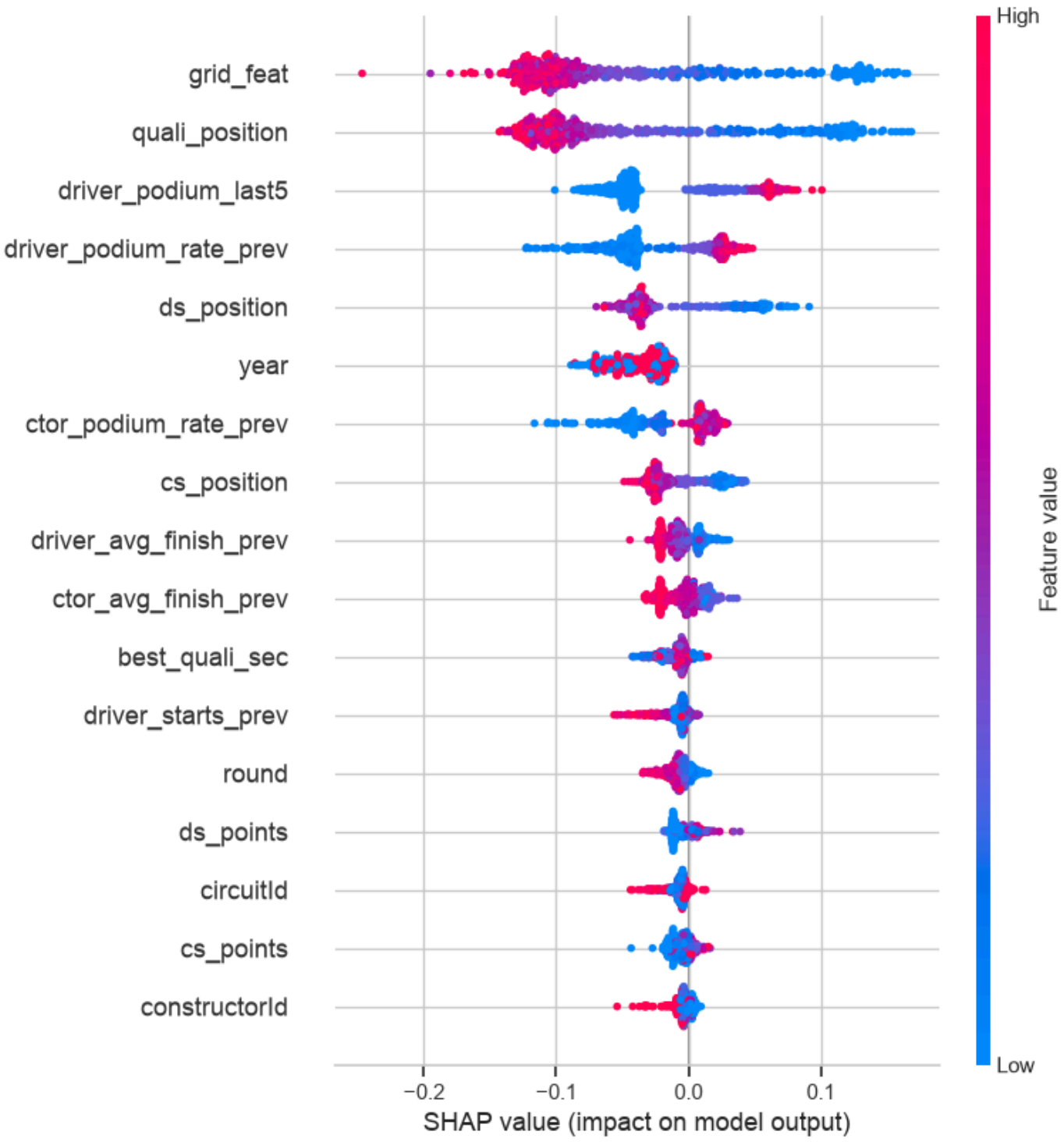


Figure 8. Global SHAP summary for the podium class.

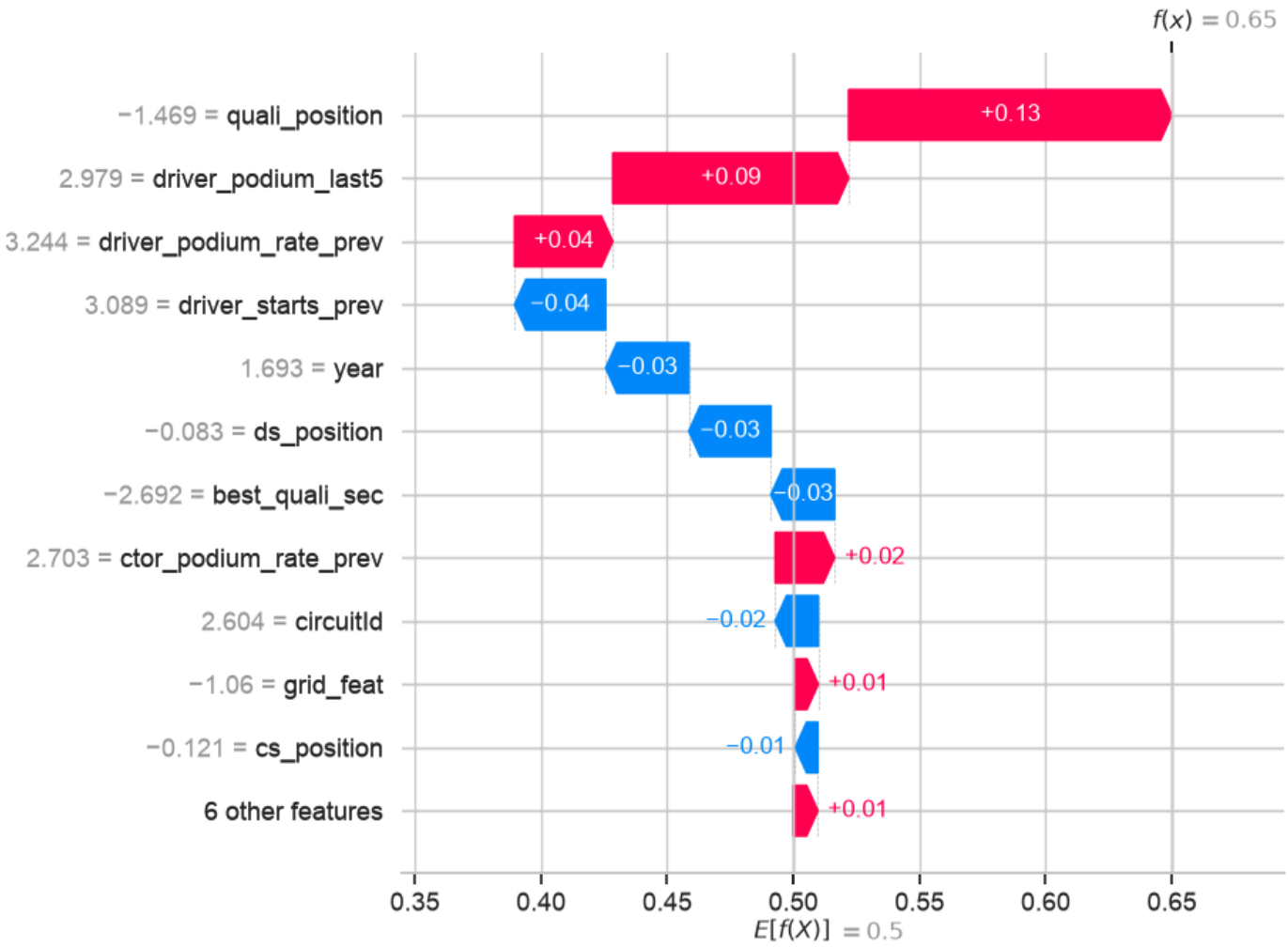


Figure 9. SHAP waterfall for one validation row.

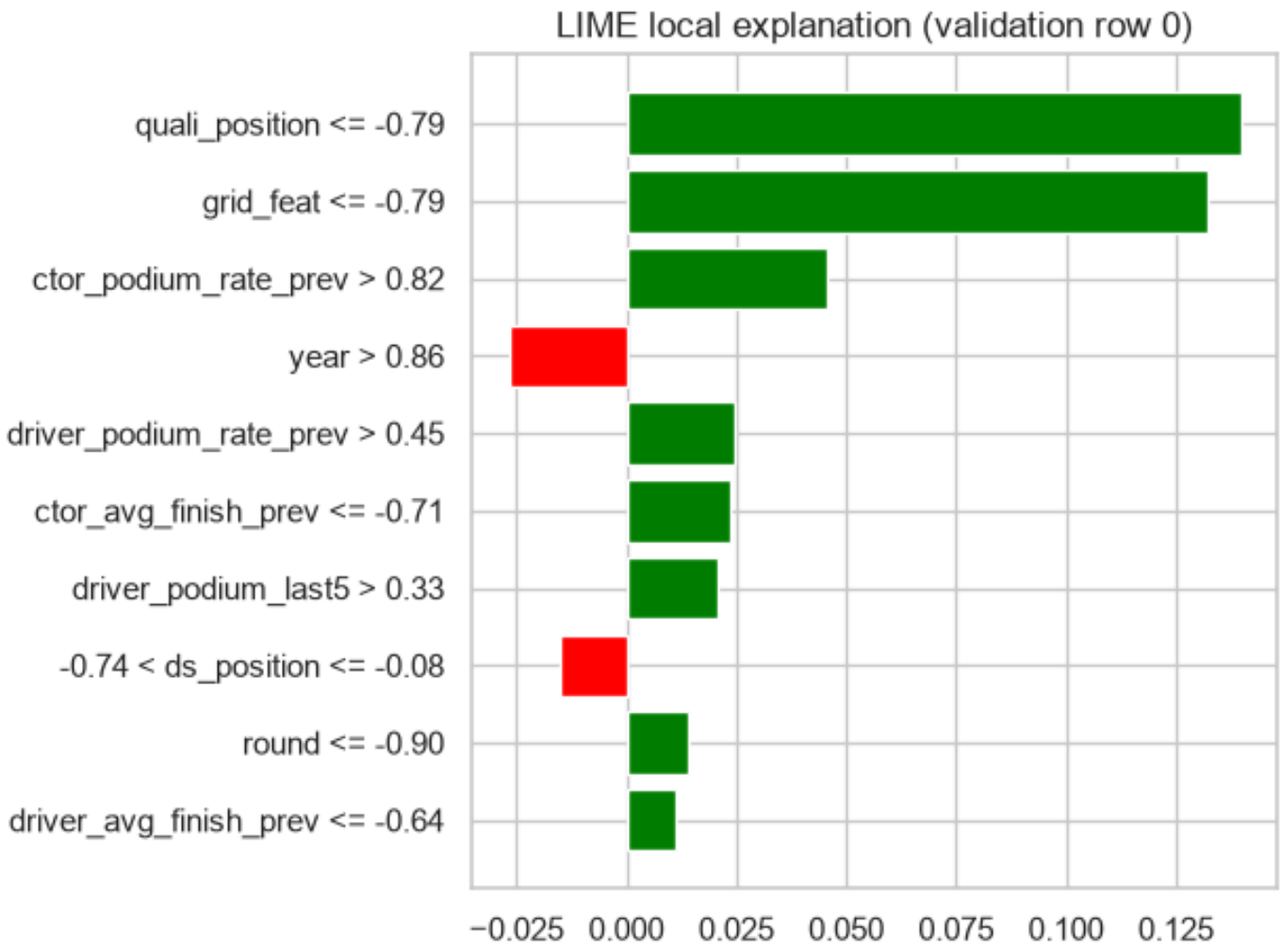


Figure 10. LIME explanation for the same style of single-row question.